Master Engineering Prompt Sequence

ANTI-GRAVITY PRO SUITE

Zero-Dataset Cold Start · Dynamic Admin Enrollment · Classical Vision (Haar + kNN) · Ethics-First Compliance

COLD-START READY | OPENCV + SCIKIT-LEARN | ADMIN & DOOR SEPARATION | LIVE FACE BLURRING

Architecture Strategy
Constructs a production-grade system starting from zero stored student photos. Includes cold-start handling, dynamic admin registration, real-time retraining, live privacy blurring, and idempotent logging.

Academic Constraints
Classical computer vision only: **Haar Cascades, Uniform LBP histograms, and distance-gated kNN (no CNNs/deep learning)**. Strict opt-in consent with automated cryptographic deletion.

Prompt 1: Project Scaffolding & Privacy-First Data Plan (M1)

Establishes the directory structure, plain-text informed consent contract, and a DoD-standard automated cryptographic data wiping script.

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ACT AS A SENIOR SYSTEMS ARCHITECT. Task: Create the repository structure and ethical data lifecycle for a classical computer vision attendance prototype. Deliverables: 1. Directory Structure: - `dataset/` (initially EMPTY; populated on-demand by admin enrollment) - `consent_forms/` (signed physical/digital PDF agreements for 4-6 volunteers) - `src/` (core modules: model_manager, admin_enroll, live_attendance, audit) - `logs/` (attendance logs, metrics, session state) 2. `consent_agreement.md`: Plain-language consent form explicitly outlining photo storage (max 10 images/person), 100x100 resolution cap, academic evaluation scope, and voluntary revocation rights. 3. `purge_data.py`: CLI script with a `--wipe` flag executing a 3-pass DoD 5220.22-M salted overwrite of all raw face crops and serialized model pickles (`model.pkl`) upon course completion. Ensure zero network telemetry, purely local sandboxed execution, and explicit terminal logging.
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Prompt 2: Zero-Dataset Model Manager & LBP Engine (M3 & M4)

Builds the feature extraction and classification core with graceful cold-start handling and empirical unknown rejection thresholding.

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ACT AS A COMPUTER VISION & MACHINE LEARNING ENGINEER. Task: Create `src/model_manager.py` handling image preprocessing, feature extraction, and dynamic kNN model compilation. Technical Requirements: 1. Cold-Start Tolerance: - If `dataset/` is empty or missing, `train_or_update_model()` must return `False` gracefully without throwing exceptions. 2. Preprocessing Pipeline (`preprocess_face`): - Input: Raw face crop (BGR or Gray). - Convert to Grayscale -> Apply CLAHE (clipLimit=2.0, tileGridSize=(8,8)) -> Resize to canonical 100x100 pixels. 3. Feature Extraction (`extract_lbp_feature`): - Compute Uniform Local Binary Patterns (LBP) with radius R=1, neighbors P=8. - Partition 100x100 image into 8x8 cells (64 blocks); calculate 59-bin normalized histogram per block; concatenate into a 3,776-dim feature vector. 4. Model Compilation & Rejection Threshold: - Fit `NearestNeighbors(n_neighbors=1, metric='euclidean')` over all enrolled subjects. - Calculate intra-class pairwise distances to compute `REJECTION_THRESHOLD = mean_dist + 2.5 * std_dist` (or calibrated fallback 0.28). - Save model, labels list, and rejection threshold to `model.pkl`.
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Prompt 3: Dynamic Admin Enrollment & Haar Tuning (M2 & M3)

Implements the live administrator enrollment tool that captures 5–10 volunteer crops and triggers immediate background model retraining.

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ACT AS A SENIOR PYTHON SOFTWARE ENGINEER. Task: Write `src/admin_enroll.py` and `src/tune_detector.py` for administrator registration. Requirements: 1. `tune_detector.py`: - Run a 3x3 grid search over `scaleFactor` in [1.05, 1.15, 1.30] and `minNeighbors` in [3, 5, 7] using `haarcascade_frontalface_default.xml`. - Output a formatted Markdown table evaluating FPS, False Positives (FP), and Miss Rate (FN). 2. `admin_enroll.py`: - Prompt admin for volunteer Student Name and Roll Number. - Open webcam stream at 30 FPS. Detect faces using tuned Haar cascade (`scaleFactor=1.15`, `minNeighbors=5`). - UI Overlay: Draw green tracking box, student name, and counter: "Captured: X/10 (Press 'c' to capture, 'q' to finish)". - On 'c': Crop detected face, run through preprocessing, and save to `dataset/<student_name>/img_<timestamp>.jpg`. - On 'q' (minimum 5 images): Automatically trigger `train_or_update_model()` from `model_manager.py` so the system is immediately trained without restarting.
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Prompt 4: Live Exam Door Inference & Privacy Blurring (M5 & M7)

Implements real-time classroom door monitoring with cold-start UI warnings, idempotent CSV logging, and instant privacy blurring.

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ACT AS A PRODUCTION SOFTWARE & PRIVACY ENGINEER. Task: Write `src/live_attendance.py` for live exam room / classroom door verification. Functional Requirements: 1. Cold-Start Safe Mode: - If `model.pkl` does not exist or contains 0 enrolled students, display a clear warning banner: "STATUS: Zero Students Enrolled. Please run admin_enroll.py" and keep camera alive without crashing. 2. Inference Pipeline: - Detect faces using Haar Cascade (`scaleFactor=1.15`, `minNeighbors=5`). Preprocess face and extract 3,776-dim LBP vector. - Query `model.pkl` for nearest neighbor distance and predicted student label. 3. ETHICS & PRIVACY MASKING (M7): - If `distance > threshold`: Face is UNKNOWN or NON-CONSENTING. - Instantly apply Gaussian Blur (`ksize=(51,51)`, `sigma=30`) over the bounding box in the live display frame. - Draw RED box labeled "Unknown (Blurred)". DO NOT save or log any image or metadata. 4. IDEMPOTENT ATTENDANCE LOGGER (M5): - If `distance <= threshold`: Display GREEN box with student name and confidence. - Check in-memory session set `marked_today`. If student not logged, append to `logs/attendance.csv`: Format: `timestamp,student_name,confidence_score,status` - Zero duplicate writes allowed per session. Atomic thread-safe file flush. Press 'q' to exit.
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Engineering Operational Guarantee

The strict decoupling between `admin_enroll.py` (write-side) and `live_attendance.py` (read-only inference) prevents model corruption during active exam sessions. Memory sets ensure duplicate registrations cannot occur within a single session.

Prompt 5: Environmental Robustness Audit Suite (M6)

Executes systematic benchmark testing across 5 lighting and occlusion conditions to assess detector and classifier degradation.

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ACT AS A QA & COMPUTER VISION VALIDATION ENGINEER. Task: Write `src/evaluate_robustness.py` to benchmark the trained pipeline across 5 controlled conditions: 1. Test Conditions (60 evaluations each): - Condition A: Diffuse nominal indoor light (500 lux). - Condition B: Low ambient light (simulated gamma = 0.40 or dark room). - Condition C: Glare / Backlight (high-intensity specular overexposure). - Condition D: Eyeglasses (frames with surface reflections). - Condition E: Face Mask (synthetic lower 45% face occlusion polygon). 2. Metrics Calculation: - Measure: Face Detection Rate (%), Recognition Accuracy (%), False Rejection Rate (FRR), and False Acceptance Rate (FAR). 3. Output: - Print a clean Markdown table comparing all 5 conditions. - Save report to `reports/robustness_metrics.md` with technical commentary explaining why Haar edge features degrade under masks.
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Prompt 6: AI Ethics, Consent & Algorithmic Bias Report (M7)

Generates the comprehensive 1-page academic ethics report covering consent agreements, demographic disparities, and misuse hazards.

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ACT AS AN AI ETHICS & GOVERNANCE SPECIALIST. Task: Generate a formal 1-page academic ethics and compliance report titled `reports/ethics_and_bias_report.md`. Required Sections: 1. Consent & Autonomy Protocol: - Detail the opt-in consent form mechanism, participant rights, zero-retention policy, and verified execution of `purge_data.py`. 2. Algorithmic Bias in Classical Vision: - Rigorous technical critique of Haar Cascades and pixel/LBP texture features across diverse Fitzpatrick skin types (I-VI). - Explain why integral image contrast differences exhibit higher miss rates on darker skin tones without CLAHE calibration. 3. Anti-Surveillance Guardrails: - Contrast passive surveillance vs. active opt-in attendance. - Explain how the real-time 51x51 Gaussian blur protects non-consenting bystanders from accidental biometric harvesting.
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Master Prompt Execution Roadmap

Phase	Prompt Target	Generated Artifact	Verification Criteria
Phase 1	Scaffolding & Consent (M1)	`consent_agreement.md`, `purge_data.py`	Zero-telemetry filesystem layout ready.
Phase 2	Cold-Start Engine (M3, M4)	`src/model_manager.py`	Zero-dataset safe; LBP 3,776-dim feature vector.
Phase 3	Admin Enrollment (M2, M3)	`src/admin_enroll.py`	Live 10-shot capture; instant model retraining.
Phase 4	Door Attendance (M5, M7)	`src/live_attendance.py`	Live unknown blur; duplicate-free CSV logger.
Phase 5	Robustness Audit (M6)	`src/evaluate_robustness.py`	5-condition stress test benchmark table.
Phase 6	Ethics Governance (M7)	`reports/ethics_and_bias_report.md`	Formal 1-page AI bias and privacy audit.